

Class 12 Homework: Pt.2 (Population Analysis) [Q13 & Q14 Boxplot]

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Section 4: Population Scale Analysis [HOMEWORK]

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale.

How many Samples do we have?

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
nrow(expr)
```

```
[1] 462
```

```
table(expr$geno)
```

```
A/A A/G G/G
108 233 121
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
expr %>%  
  group_by(geno) %>%  
  summarise(median_expression = median(exp))
```

```
# A tibble: 3 x 2  
  geno median_expression  
  <chr>          <dbl>  
1 A/A             31.2  
2 A/G             25.1  
3 G/G             20.1
```

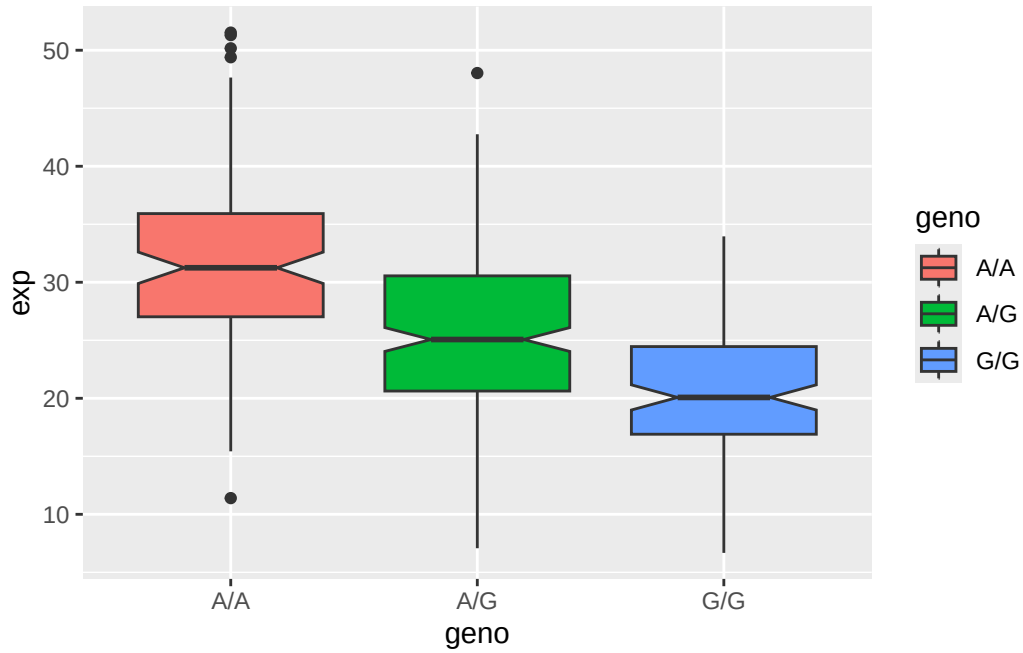
The total sample size is 462. The sample size of G|genotype A|A is 108, A|G with 233, and G|G with 121. The median expression level for A|A is 31.25, A|G at 25.06, and G|G at 20.07.

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)
```

Let's make a boxplot:

```
ggplot(expr) + aes(geno, exp, fill=geno) +  
  geom_boxplot(notch=TRUE)
```



Based on the boxplot, its clear that having a G|G in this location has significantly reduced expression compared to A|A in this gene. This suggests that the SNP affects ORMDL3 expression, since different genotypes are associated with distinct expression levels, consistent with a regulatory effect.